

EDUCATION

University of Missouri - Columbia (2008-2011)
Tianjin University of Technology (2003-2007, China)

M.A. in Statistics
B.S. in Cost Engineering

Language and Cloud Environment

- **Programming:** Python, SQL
 - **Predictive modeling:** Xgobst, Random forest, KNN, Kmeans, time serise, Prophet, Logistic
 - **Embedding models:** Clip, Sbert, Faiss
 - **Cloud Platforms:** AWS (IAM Roles, S3, DynamoDB, Personalize, Step Functions, etc.)
Azure DevOps (Terraform, Pipeline Orchestration)
Databrick: Pyspark/python notebook, UC catalog, DLT pipeline, jobs
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CAREER EXPERIENCE

❖ Sr. Machine Learning Engineer | CoStar Group (Homes.com) Dec 2023 – Jun 2025

- Developed and implemented a computer vision system utilizing OpenAI's CLIP model to generate embeddings for condominium building images and applied zero-shot classification techniques to accurately label these images without additional training data.
- Implemented SBERT to create embeddings for textual data, enhancing the performance of recommendation systems and similarity searches across various data types.
- Fine-tuned Bert Models (LLMs) to enhance home search functionality by enabling feature-based search inputs, bypassing traditional filters.
- Built and maintained pipeline orchestration for model deployment and data ingestion in retargeting projects across production and pre-production environments using AWS and Databricks.
- Developed innovative solutions to address model gaps, optimize AWS costs, and evaluate newly released AWS models.
- Updated and designed streaming DLT pipelines and corresponding dashboards to reflect with new features added into homepage, set up put events to feed real-time events to the recommendation system to close the feedback loop.

❖ Machine Learning Engineer II | Glidewell Jan 2021 – Dec 2023

- Designed and deployed a RAG-based GenAI chatbot using AWS Kendra for web content ingestion, Bedrock for LLM inference to enhance internal customer service efficiency.
- Implemented XGBoost for churn prediction, reducing customer attrition by 12% through personalized engagement strategies.
- Built a customer segmentation model to classify customers into tiers and designed tailored business use cases for each segment to drive targeted engagement and strategy.
- Collaborated with MLOps to implement CI/CD pipelines, successfully deploying models in AWS environments.

❖ Sr. Data Scientist | New Century Health Apr 2019 – Dec 2020

- Built predictive models for portfolio cost projections, driving strategic decisions.

- Analyzed healthcare data, including drug code, diagnose code, CPT code v.s. cost, to create actionable insights for diagnosis and treatment.

❖ **Sr. Data Scientist, Commercial Marketing Analytic | Aetna** **Aug 2017 – Apr 2019**

- Developed machine learning models that drove \$11.7M in incremental revenue across B2B and B2C channels.
- Conducted segmentation analyses and collaborated on A/B testing to optimize marketing strategies.

❖ **Data Scientist Sr. Advisor, Decision Science | Dell Financial Service** **Sep 2013 – Jul 2017**

- Built credit risk models, improving capture rates and reducing credit losses by 12%.
- Conducted marketing campaign optimizations through segmentation and profitability analysis.

❖ **Statistical Analyst | Javelin Marketing Group** **Aug 2011 – Sep 2013**